Pearls AQI Predictor

Internship Project Final Report

Karachi Air Quality Forecasting for 24, 48, and 72 Hours

Final submission: June 7, 2026

1. Introduction

Pearls AQI Predictor is an end-to-end air quality forecasting system for Karachi, Pakistan. The project predicts Air Quality Index values for the next 24, 48, and 72 hours using recent pollutant, weather, and time-based features.

The system includes data fetching, feature engineering, model training, prediction generation, model explanation, dashboard visualization, and GitHub Actions automation.

2. Problem Statement

Air pollution can change quickly due to weather conditions, pollutant concentration, and time-based activity patterns. The objective of this project is to build a forecasting pipeline that can estimate near-future AQI and present the result in a simple dashboard for monitoring and decision support.

3. Data Source

The project uses Open-Meteo APIs for Karachi:

- Weather archive data
- Air quality and pollutant data

The final dashboard snapshot contains 2,184 hourly rows from 08 Mar 2026 to 06 Jun 2026.

Main variables:

- US AQI
- PM2.5
- PM10
- Carbon monoxide
- Nitrogen dioxide
- Sulphur dioxide
- Ozone
- Temperature
- Relative humidity
- Wind speed and direction
- Pressure
- Cloud cover

4. System Architecture

```
Open-Meteo APIs
|
v
src/fetch_data.py
|
v
Raw CSV data
|
v
src/features.py
|
v
Training feature dataset
|
+--> src/train.py --> trained models + metrics
|
```

```
+--> src/explain.py --> feature importance
|
+--> src/predict.py --> latest 3-day AQI forecasts
|
v
Streamlit dashboard
```

Automation is handled through GitHub Actions:

- Hourly prediction workflow
- Daily model training workflow

5. Feature Engineering

The feature engineering pipeline creates:

- Time features: hour, day of week, day of month, month, weekend flag
- Lag features: 1h, 3h, 6h, 12h, 24h, 48h, 72h
- Rolling statistics: 3h, 6h, 12h, 24h, 48h, 72h rolling mean and standard deviation
- Change features: AQI and pollutant changes over time
- Forecast targets:
- `target_aqi_24h`
- `target_aqi_48h`
- `target_aqi_72h`

The processed training dataset contains 2,040 clean rows and 188 columns.

6. Exploratory Data Analysis

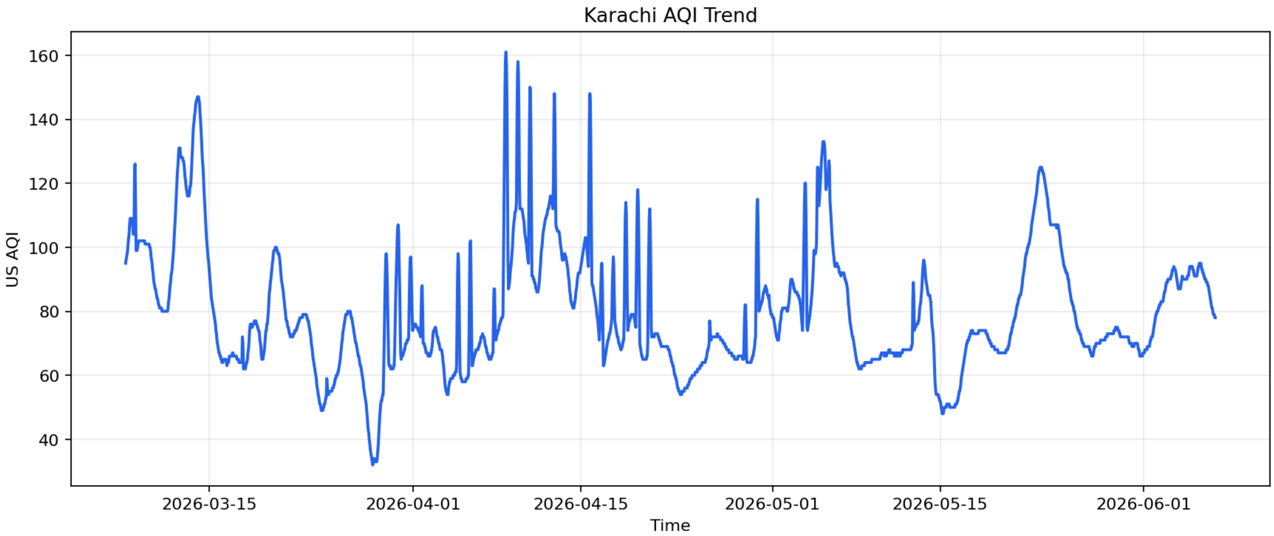
EDA found:

- Rows analyzed: 2,184
- Missing values: 0
- Mean AQI: 80.01
- Minimum AQI: 32
- Maximum AQI: 161

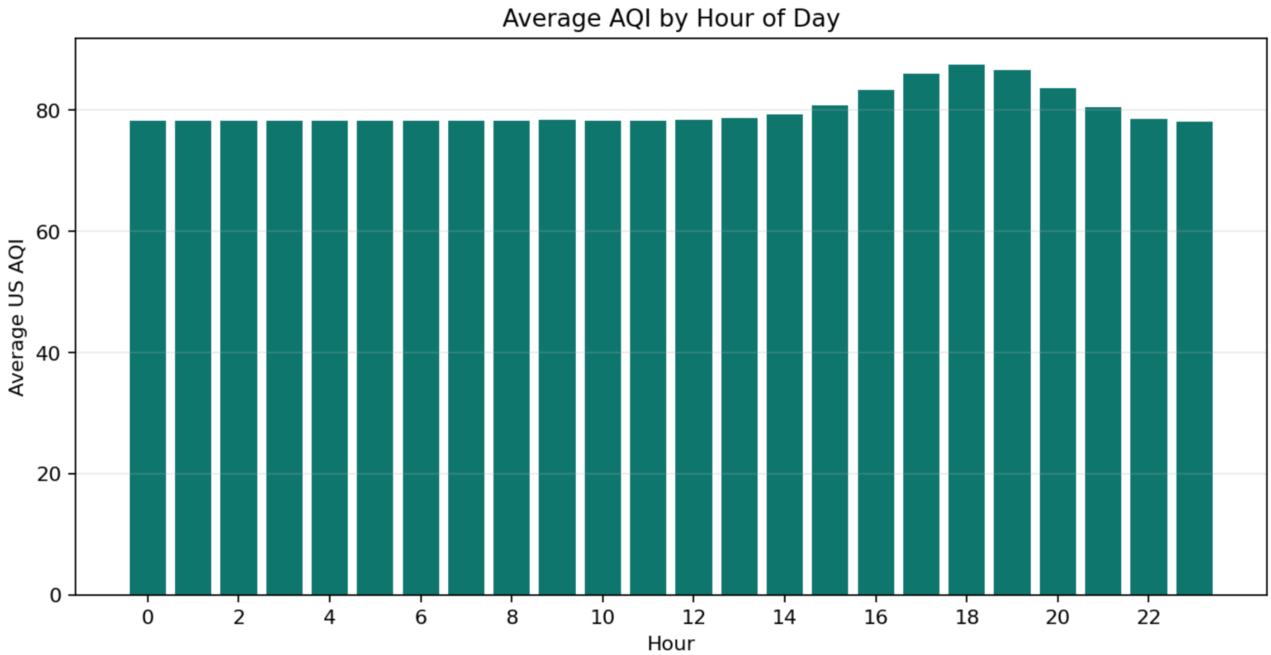
AQI category counts:

Category	Count
Good	57
Moderate	1,807
Unhealthy for Sensitive Groups	314
Unhealthy	6
Very Unhealthy	0
Hazardous	0

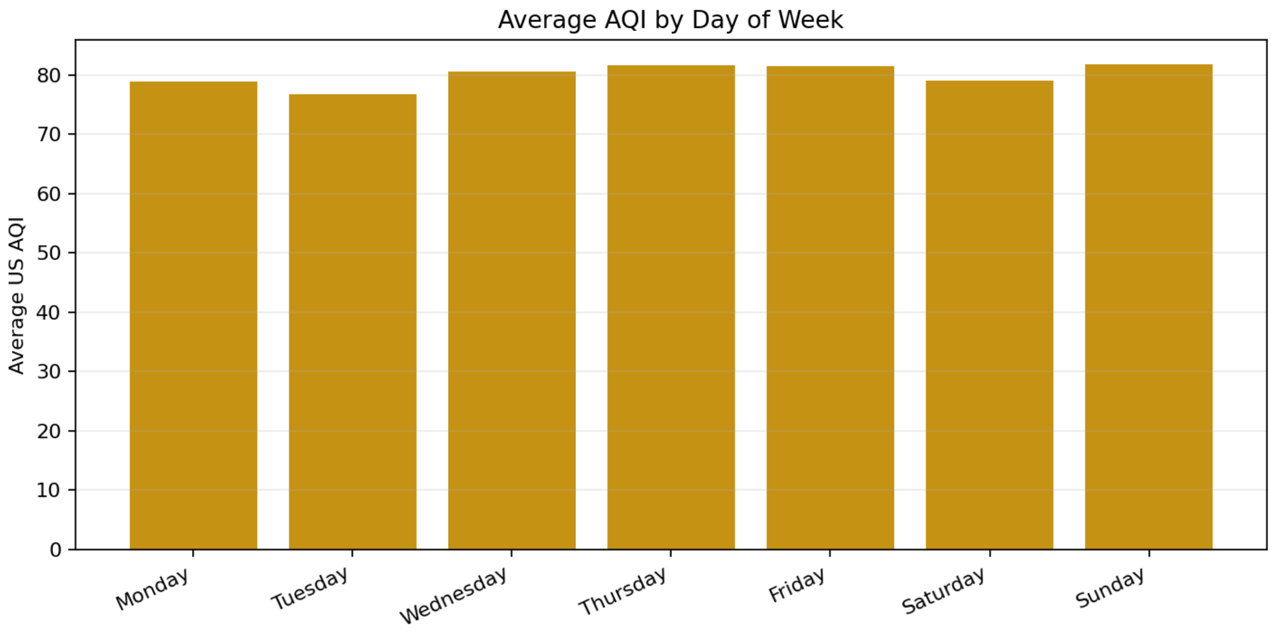
Generated EDA figures:



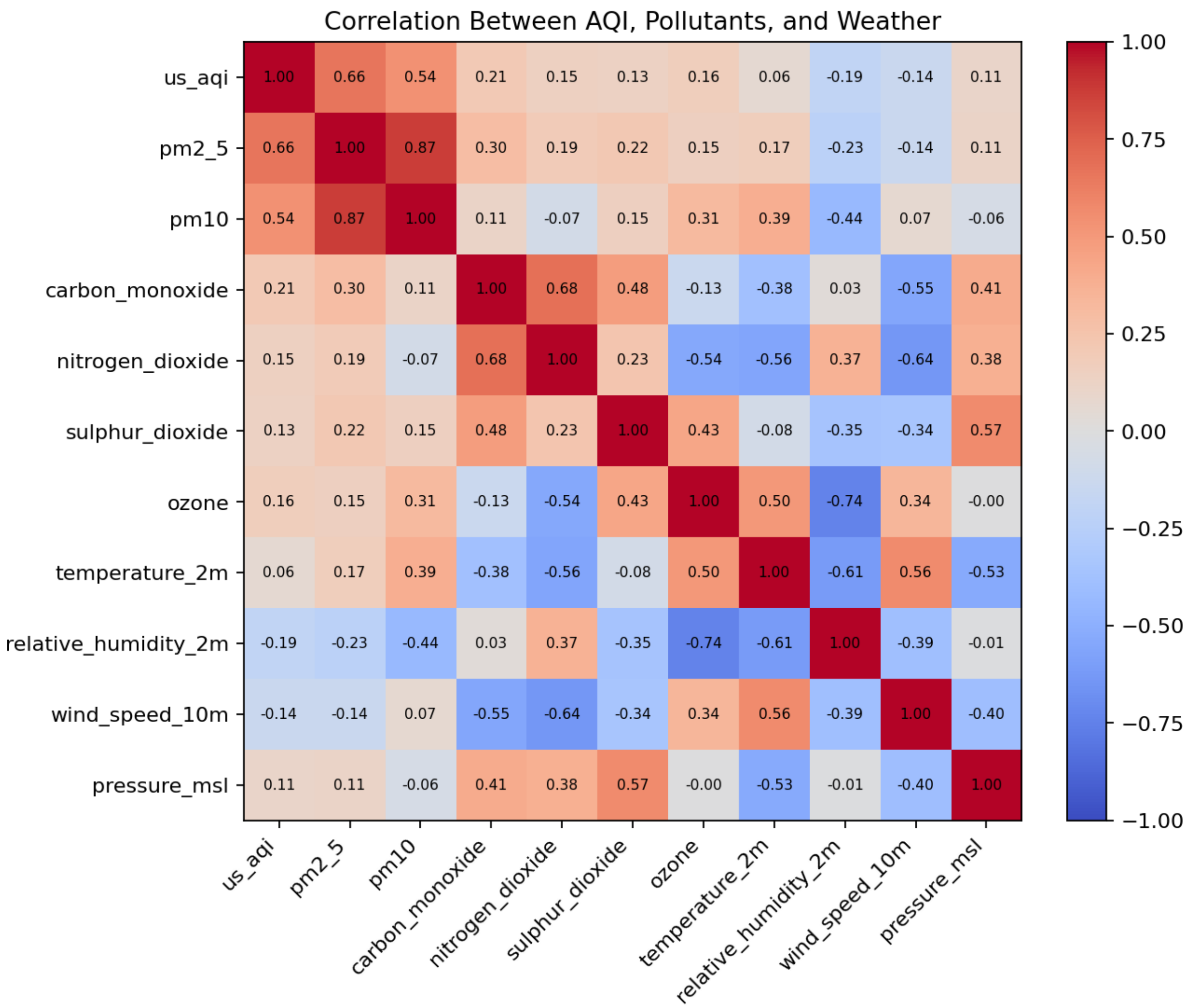
AQI trend



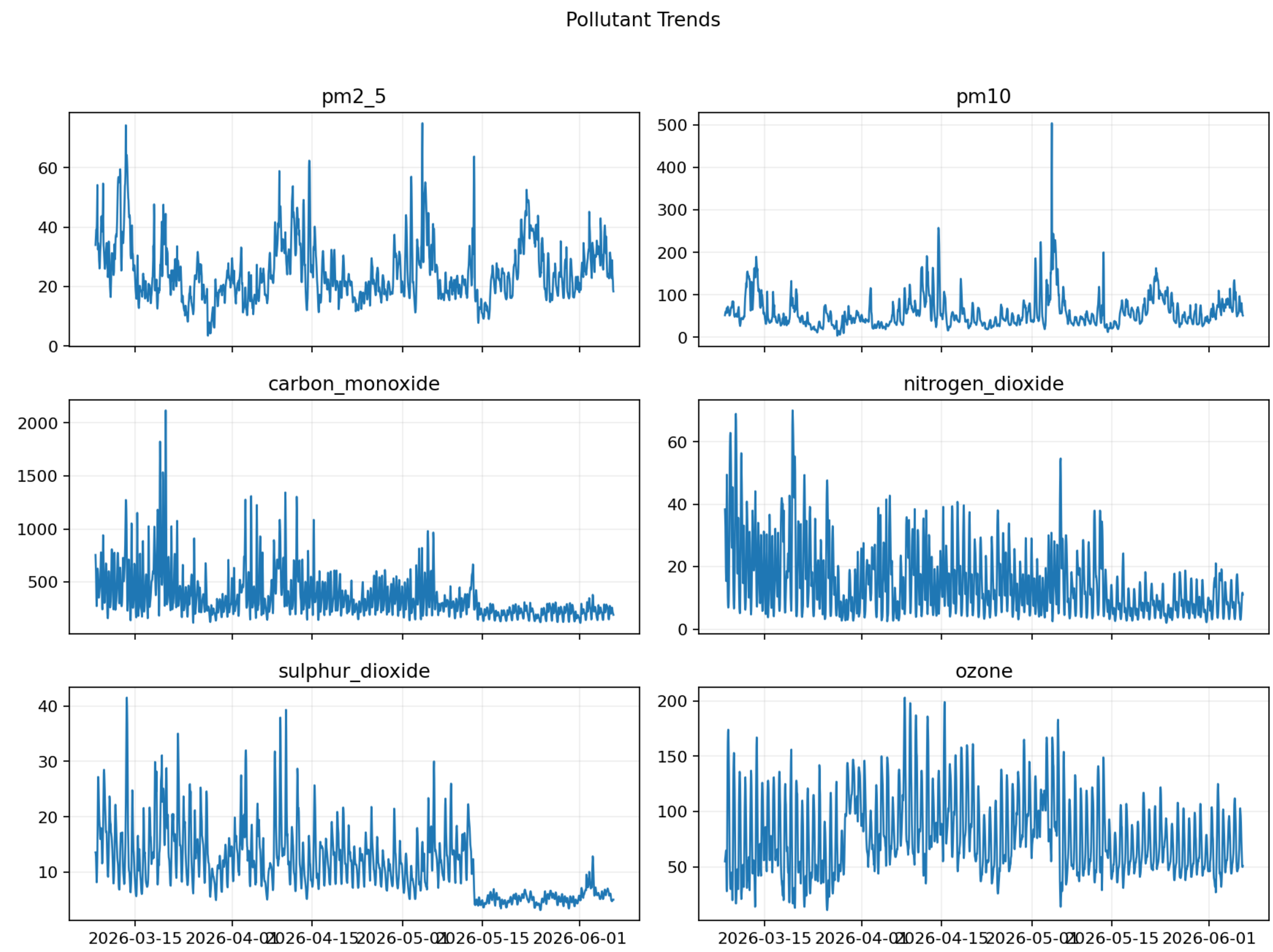
Hourly AQI pattern



Daily AQI pattern



Pollutant/weather correlation



Pollutant trends

7. Models Used

Three model approaches were used:

Model	Purpose
Current AQI baseline	Benchmark that assumes future AQI equals current AQI
Ridge Regression	Linear regularized model
Random Forest Regressor	Non-linear tree-based model

8. Model Evaluation

Evaluation used a time-based train/test split to avoid future leakage. Metrics used:

- MAE
- RMSE
- R2

Results:

Target	Model	MAE	RMSE	R2
24h	Current AQI baseline	8.90	11.30	0.492
24h	Random Forest	9.20	12.16	0.412

Target	Model	MAE	RMSE	R2
24h	Ridge	10.34	13.72	0.251
48h	Random Forest	12.77	17.04	-0.191
48h	Current AQI baseline	14.78	19.54	-0.566
48h	Ridge	16.33	20.39	-0.706
72h	Random Forest	14.87	20.55	-0.841
72h	Current AQI baseline	19.09	24.89	-1.701
72h	Ridge	30.86	34.68	-4.244

The current-AQI baseline performed best at 24 hours. Random Forest performed best among the tested models at 48 and 72 hours. The negative R2 values at longer horizons indicate that the current 90-day dataset is not sufficient for strong long-range forecasting.

9. Latest Forecast Snapshot

The final forecast snapshot was generated from the 06 Jun 2026, 11:00 PM observation.

Horizon	Forecast Time	Model	Current AQI	Predicted AQI	Category
24h	07 Jun 2026, 11:00 PM	Current AQI baseline	78	78.00	Moderate
48h	08 Jun 2026, 11:00 PM	Random Forest	78	72.35	Moderate
72h	09 Jun 2026, 11:00 PM	Random Forest	78	92.03	Moderate

10. Explainability

The project includes `src/explain.py`, which generates feature importance values for forecast models.

For the local environment, Random Forest feature importance is used. The script is also written to use SHAP when SHAP is installed and available.

The selected 24h model is the baseline and does not expose tree-based feature importance. Top drivers for the 48h Random Forest model include:

- Day of month
- Wind speed rolling mean over 48h
- PM2.5 rolling mean over 6h
- Temperature rolling mean over 48h
- PM10 rolling mean over 72h

These values are displayed in the dashboard under the Forecast drivers section.

11. Dashboard

The Streamlit dashboard includes:

- Current AQI and AQI category
- Pollutant profile
- 24h, 48h, and 72h forecasts
- Observed and forecast AQI charts
- Model evaluation
- Feature importance
- Automated pipeline overview

Live dashboard:

https://pearls-aqi-predictor-8eluvoxqdhfnpt5iyzfogl.streamlit.app

12. Automation

The project includes two GitHub Actions workflows:

Workflow	Purpose
Hourly AQI Predictions	Fetch data, build features, create fallback models if needed, explain models, and generate predictions
Daily AQI Model Training	Fetch data, build features, train models, explain models, and generate predictions

Both workflows were manually tested successfully. Artifacts are uploaded after workflow runs.

13. Limitations

- The current model uses around 3 months of data.
- Longer historical data would improve seasonal learning.
- The deployed dashboard uses a committed data snapshot for easy public demo access.
- A production version should use a feature store and model registry.
- The 48h and 72h forecasts have negative R2 values on the latest test window.

14. Future Work

- Add 1-2 years of historical training data.
- Integrate Hopsworks Feature Store.
- Store trained models in a proper model registry.
- Add full SHAP visualizations.
- Add multi-city support.
- Improve model selection with XGBoost or LightGBM.
- Deploy an automated data refresh backend for Streamlit Cloud.

15. Conclusion

The project successfully implements an end-to-end AQI prediction system with data ingestion, feature engineering, model training, prediction generation, explainability, automation, and dashboard deployment. The latest evaluation demonstrates that model selection changes by horizon and data window: the baseline is strongest at 24 hours, while

Random Forest is selected at 48 and 72 hours. Longer historical coverage is the most important next step for improving forecast reliability.